



# Optimizing Solar Panel Integration: Rooftop and Ground-Level Strategies for Sustainable Schools

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## Introduction

High school students rely on consistent AC power to support critical classroom technologies and laboratory equipment, forming the backbone of an effective learning environment. Recognizing the importance of reliable energy, this project examines the potential of sustainable solar panel systems to power high schools. By integrating core mechanical, electrical, and environmental engineering principles, the initiative aims to develop renewable energy solutions that are both practical and resilient.

As part of a community college electrical engineering course, students conducted evaluations at various high school locations to assess how real-world conditions—such as sunlight exposure and optimal panel placement—impact energy production. The research involved comparing different configurations, including rooftop and ground-level systems, to identify the most effective strategies for maximizing energy output.

A detailed case study at President William McKinley High School's C-building provided focused insights into practical installation challenges and design considerations. The project highlights how embracing renewable energy can reduce operating costs and lower carbon footprints while enhancing overall energy efficiency in school infrastructures, ultimately contributing to more sustainable community spaces and preparing future engineers for real-world challenges.

## Design

C-building for President William McKinley High School served as a proof of concept for this project. During the redesign process in Aladdin, integrating the existing small rooftop HVAC installations introduced notable challenges to accurately capturing the building's geometry. Consequently, I adopted a more streamlined design approach that preserved the essential character of the structure while accommodating these complexities.

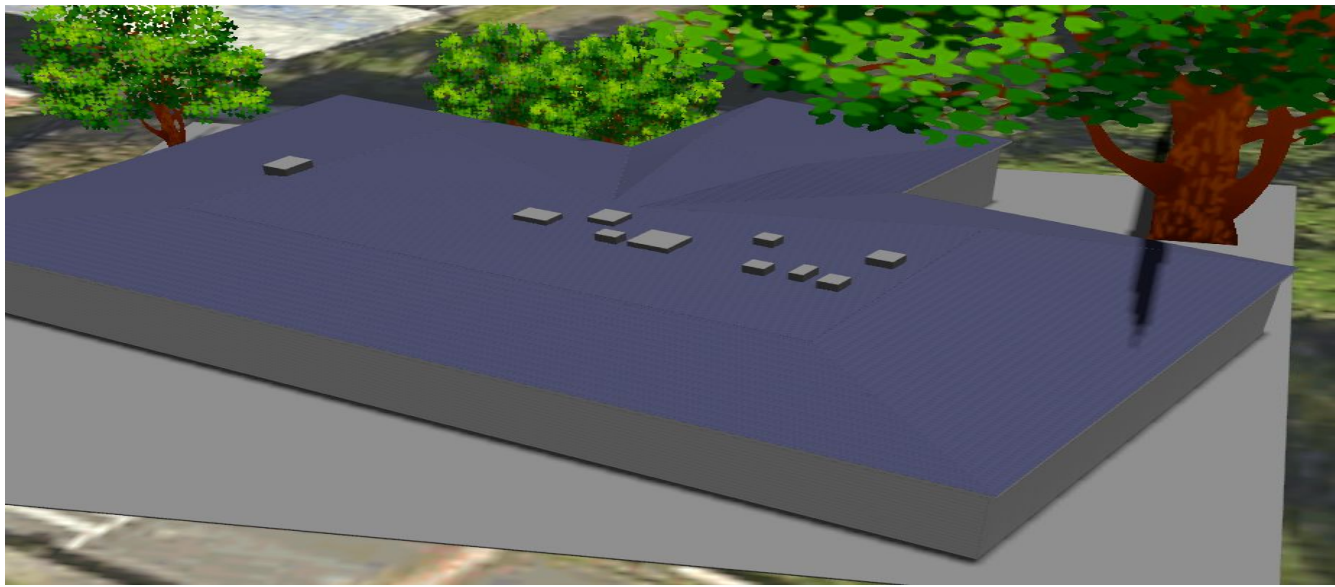


Fig. 1: This is the final design of C-Building showcasing multiple distinct roof features that helped establish the ideal positioning of the solar panels

Once the redesign was complete, the next step involved placing the solar panels to maximize coverage and account for the updated roof features. I ultimately distributed them in four distinct sections—left, right, front, and back—ensuring each array would capture ample sunlight while accommodating the building's revised layout.

## Testing

The key objective of the design is to verify that it meets its intended goals. My aim—and I believe this is a common aspiration—is for the system to eventually pay for itself while generating more energy than it consumes. To evaluate this, I utilized the Aladdin site to run a simulation on my final design.

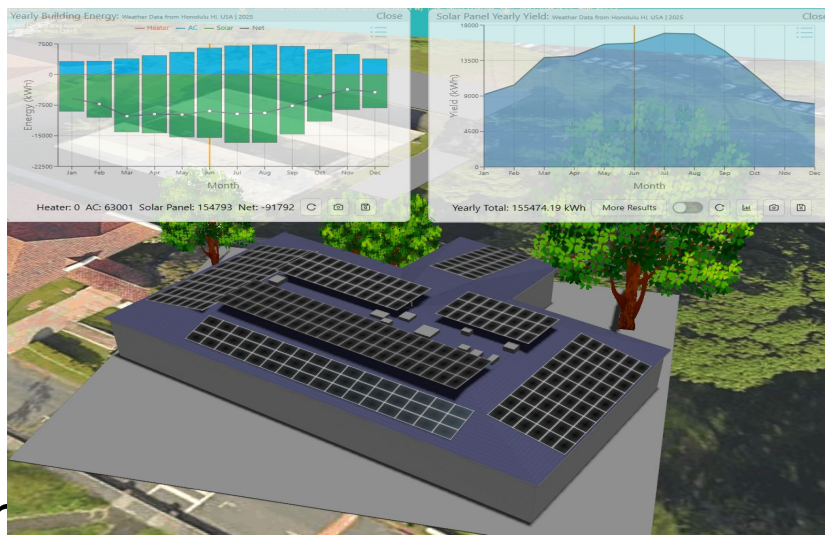


Fig. 2: Final Design showing yearly energy generated by the building and yearly energy consumption.

## Conclusion and Future Work

Integrating rooftop and ground level solar systems is essential for advancing renewable energy and reducing reliance on external sources. Rooftop arrays transform underutilized space on buildings into local power generators, lowering utility costs which the benefits are valuable in remote areas.

For high schools, a steady supply of AC power is crucial to maintain a conducive learning environment. Consistent electricity supports efficient air conditioning and modern classroom technology, which in turn enhances student productivity and overall comfort. Solar installations help ensure that educational facilities operate reliable and economically.

At the same time, large-scale ground level solar farms offer centralized power production to meet increasing energy demands. These expansive systems contribute to broader clean energy goals by foster environmental sustainability, strengthening local economies, and promoting energy independence. Together, these solar approaches provide a balanced, scalable model for sustainable growth while supporting the critical needs of community institutions.

## References

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